

SACUIENI, Romania

WMO: 150420

Lat: 47.3442N

Lon: 22.0944E

Elev: 412

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	16.7	6.0	7.6	14.2	11.7	10.2	19.2	23.2	41.7	20.4	40.5	3.0	50	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.9	90.7	71.3	87.9	70.4	85.0	69.0	73.8	86.6	72.0	84.1	70.5	82.0	4.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	111.4	79.8	68.0	104.4	78.1	66.4	98.9	76.2	37.6	86.1	35.9	84.7	34.7	82.0	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.3	16.7	14.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	30.2	36.4	43.6	53.2	62.4	68.3	70.2	71.1	62.0	52.5	41.9	34.1
	DBStd	16.06	8.29	8.56	8.26	8.36	6.44	6.01	5.71	6.33	7.00	8.10	9.20	8.54
	HDD50	2138	615	382	226	65	4	0	0	0	2	69	276	499
	HDD65	5281	1080	800	663	361	125	34	17	17	141	391	694	958
	CDD50	2955	0	2	28	159	387	550	627	656	361	147	32	6
	CDD65	624	0	0	0	6	43	134	179	207	51	4	0	0
	CDH74	5994	0	0	1	55	370	1196	1654	2171	498	48	1	0
	CDH80	2079	0	0	0	4	55	383	569	918	150	0	0	0
Wind	WSAvg	5.1	4.8	6.1	6.1	6.3	5.8	4.5	4.2	3.7	4.1	4.4	5.1	5.7
Precipitation	PrecAvg	23.2	1.2	1.2	1.3	1.7	2.2	3.1	2.8	1.9	2.2	1.9	1.9	1.9
	PrecMax	33.6	2.5	3.7	5.5	2.8	5.0	6.0	5.0	5.6	6.4	4.3	3.3	4.3
	PrecMin	15.1	0.4	0.1	0.3	0.2	0.4	0.5	0.4	0.1	0.4	0.1	0.0	0.1
	PrecStd	5.1	0.6	0.8	1.3	0.7	1.1	1.6	1.3	1.4	1.4	1.2	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	58.5	71.2	79.1	84.7	91.5	92.6	96.2	89.4	78.2	67.2	59.8
		MCWB	47.3	49.8	55.5	61.3	65.7	73.1	71.8	73.5	67.6	61.4	56.2	51.5
	2%	DB	48.0	54.1	64.9	74.8	81.4	88.2	89.2	91.9	83.9	75.0	62.5	53.7
		MCWB	44.5	48.0	52.6	59.1	64.9	71.6	71.5	70.9	65.4	60.2	54.4	47.9
	5%	DB	44.7	51.5	59.9	70.6	78.3	84.7	86.0	88.5	80.1	70.9	58.8	49.5
		MCWB	41.7	46.3	50.1	57.0	64.2	70.0	70.4	70.2	64.6	58.6	52.8	45.2
	10%	DB	41.5	48.5	55.8	67.1	75.2	81.0	82.9	85.6	75.2	66.3	54.8	44.8
		MCWB	39.2	44.1	47.4	54.9	62.0	68.2	68.9	68.8	63.1	56.6	50.6	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	51.7	57.3	63.2	68.3	76.4	76.2	75.3	69.2	64.3	57.0	52.1
		MCDB	51.7	57.4	69.3	75.6	79.6	88.6	89.2	92.5	82.5	71.0	64.5	59.0
	2%	WB	45.2	48.8	53.7	60.5	66.6	73.4	73.4	73.3	67.2	61.2	55.3	48.7
		MCDB	48.6	53.5	63.0	72.4	78.1	85.0	83.9	88.4	80.1	73.1	62.0	53.2
	5%	WB	41.8	46.7	50.8	58.3	64.9	71.3	71.7	71.6	65.6	59.6	53.4	45.4
		MCDB	44.2	51.1	58.6	69.2	75.3	81.8	82.4	85.8	77.4	69.7	58.5	49.2
	10%	WB	39.0	44.6	48.4	56.1	63.4	69.4	70.2	70.0	63.9	57.3	51.4	42.4
		MCDB	41.1	48.1	54.6	65.7	72.7	79.3	80.5	83.1	73.8	65.5	55.3	44.9
Mean Daily Temperature Range	5% DB	MDBR	9.5	11.8	15.9	19.1	19.6	19.4	20.6	21.9	19.1	19.0	12.5	8.6
		MCDBR	11.4	15.7	21.5	23.7	23.2	23.7	24.6	26.4	26.6	26.0	16.6	12.3
		MCWBR	7.9	10.8	12.5	12.0	10.6	10.5	10.2	9.7	11.2	13.9	9.3	8.9
	5% WB	MCDBR	10.2	15.1	20.5	22.2	21.5	22.1	21.7	24.8	24.5	24.8	15.2	11.5
MCWBR		7.8	10.7	12.5	11.5	10.5	10.5	9.6	9.9	10.3	13.2	8.8	8.4	
Clear-Sky Solar Irradiance	taub	0.332	0.358	0.397	0.450	0.434	0.434	0.453	0.451	0.417	0.385	0.358	0.333	
	taud	2.314	2.272	2.201	2.078	2.193	2.224	2.180	2.181	2.242	2.309	2.331	2.344	
	Ebn at Noon	232	251	258	254	264	264	257	252	249	238	220	217	
	Edh at Noon	24	31	39	48	44	43	45	43	37	30	24	21	
All-Sky Solar Radiation	RadAvg	331	596	996	1431	1747	1899	1863	1702	1188	771	423	251	
	RadStd	45	92	141	143	171	126	160	152	138	91	51	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	+1.24	N/A	N/A	+1.87	N/A	N/A	-229	-349	+203	+103

Nomenclature: See separate page